Use python opencv to return the smallest circumscribed rectangle of the point set cnt, the function used is cv2.minAreaRect(cnt), cnt is the point set a vector of the minimum circumstantial rectangle required, and the number of points in this point set is indefinite.

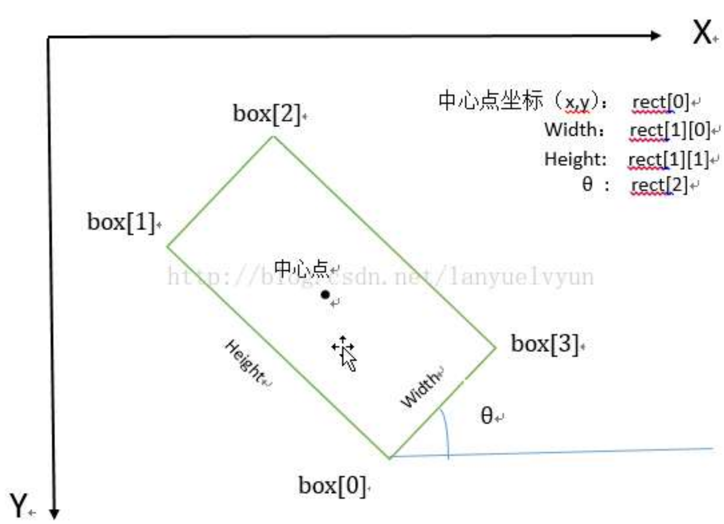
For example: draw the smallest circumscribed rectangle of an arbitrary quadrilateral, where cnt represents the 4 vertex coordinates of the quadrilateral (t are 4 points in the point set)

cnt = np.array([[x1,y1],[x2,y2],[x3,y3],[x4,y4]]) # must be in the form of an array array

rect = cv2.minAreaRect(cnt) # Get the smallest circumscribed rectangle (center(x,y), (width, height), rotation angle)
box = cv2.cv.BoxPoints(rect) # cv2.boxPoints(rect) for OpenCV 3.x Get the 4 vertices of the smallest circumscribed rectangle
box = np.int0(box)

The function cv2.minAreaRect() returns a Box2D structure rect: (center of the smallest circumscribed rectangle (x,y), (width, height), rotation angle). But to draw this rectangle, we need the 4 vertex coordinates of the box, obtained by the function cv2.cv.BoxPoints(), box:[[x0,y0], [x1,y1], [x2,y2], [x3,y3]

The correspondence of the 4 vertex order, center coordinates, width, height, and rotation angle (in the form of degrees, not radians) of the minimum circumscribed rectangle is as follows:



Note: The rotation angle θ is the angle between the horizontal axis (x-axis) rotating counterclockwise and the first edge of the rectangle it touches. And tl length of this side is width, and the length of the other side is height. That is, here, width and height are not defined by length.

In OpenCV, the origin of the coordinate system is in the upper left corner, with a negative counterclockwise rotation angle and a positive clockwise rotatic angle relative to the x-axis. Here, θ ∈ (-90 degrees, 0].

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